Project 2 Part B - Statistical and Ethical Analysis of NASA Challenger O-Ring Failure

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1. Logistic Regression Summary and Predictions

We fit a logistic regression model using Damaged (0/1) as the outcome and air temperature (Temp) as the predictor. The fitted model is:

P(Damaged = 1) = 1 / [1 + e^-(5.0686 - 0.1153 * Temp)]

The negative temperature coefficient indicates that colder launch temperatures are associated with higher probabilities of O-ring damage. Figure 1 shows this inverse relationship: as temperature decreases, predicted damage risk rises. The model's predicted probabilities at representative temperatures are:

- At 50°F: predicted probability of damage = 0.3321
- At 40°F: predicted probability of damage = 0.6118
- At 30°F: predicted probability of damage = 0.8332

These results demonstrate that the probability of O-ring damage increases rapidly as temperature decreases, particularly below 50°F, indicating a nonlinear escalation in risk under colder conditions. Because 40°F and 30°F fall below most observed temperatures in the data, these predictions are extrapolations and should be interpreted with caution, as they fall outside the primary range of observed data and therefore carry greater uncertainty. However, they still reinforce the strong cold-weather risk captured by the model.

2. Ethical Principle: Professional Judgment vs Organizational Pressure

This principle means that a data scientist's first obligation is to truth, evidence, and the protection of public safety, not schedule pressure, political pressure, or management preferences. Professional judgment

includes communicating uncertainty, highlighting model limitations, and refusing to present weak evidence. If an organization wants a comforting answer, the ethical responsibility is not to provide comfort but to provide an accurate risk assessment, even when that is inconvenient.

Consequences can be severe when this principle is violated. Immediate harms can include unsafe decisions, avoidable accidents, and loss of life. Long-term harms include erosion of trust in technical teams, damaged organizational credibility, and a culture in which analysts learn to optimize for approval rather than validity. Once that culture forms, warning signals are even more likely to be ignored in future high-stakes decision making. In practice, this requires analysts to clearly communicate risk levels and uncertainty, even when conclusions may conflict with organizational expectations.

3. Risk Framing, Bias, and Ethical Decision-Making

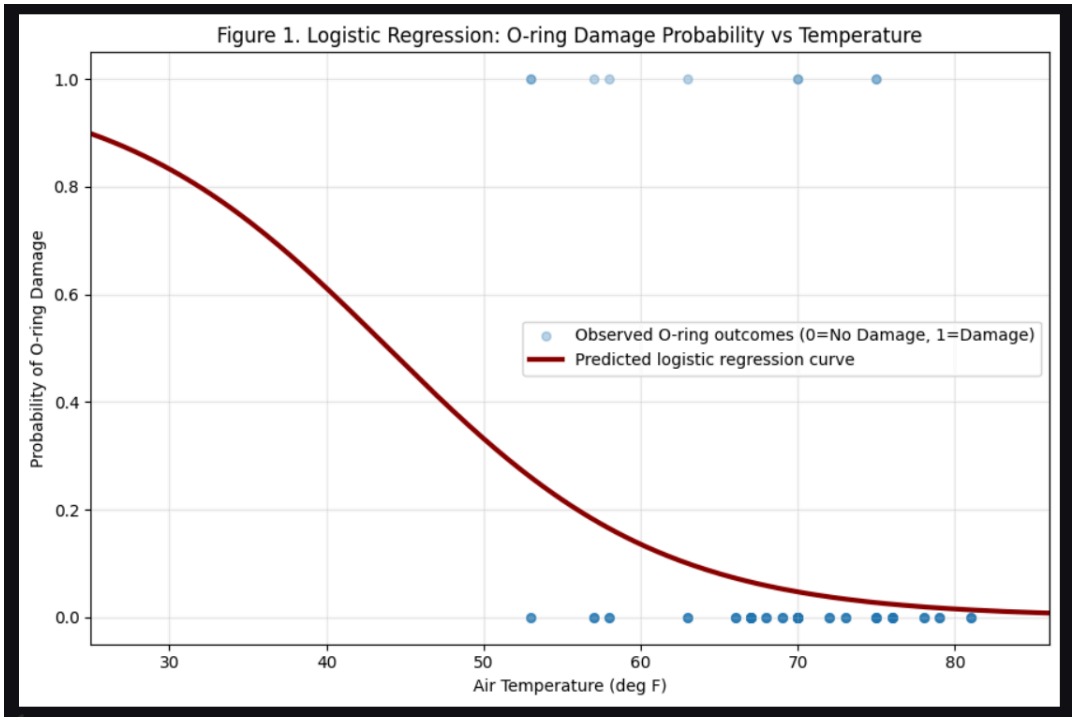
Asking analysts to prove a launch is safe can create an unethical burden of proof because it frames safety as the default and danger as an exception. In high-risk systems, the safer ethical approach is to flip the question: if there is not strong evidence that conditions are safe, leaders should assume the risk is still unresolved. The “prove it is safe” framing can pressure teams to search for supporting evidence while discounting contradictory data, especially when there is limited historical data or conditions are outside the tested range.

This framing directly fuels confirmation bias: people selectively focus on results that support launch readiness and rationalize warning signs as noise. Ethical risk communication requires presenting both what is known and what remains uncertain, including worst-case outcomes and confidence limits. Data scientists have a duty to communicate when evidence is insufficient, and leaders have a duty to treat that uncertainty as actionable information rather than an obstacle to be managed away. This is precisely what happened with Challenger: engineers at Morton Thiokol explicitly raised O-ring concerns the night before launch and were overruled by managers who treated their uncertainty as an obstacle rather than a stop signal.

4. Reliability Standards in High-Risk Engineering Systems

In high-risk missions, technology should be proven by robust performance under realistic worst-case conditions, not merely by average performance. The threshold should be high enough that failure probabilities are low across the full operational envelope, including edge cases such as extreme temperatures, unusual loads, or degraded components. This requires conservative testing, independent review, and explicit treatment of uncertainty before approving mission-critical use.

Ethically, the threshold is guided by nonmaleficence (avoiding preventable harm), duty of care to astronauts and ground personnel, and accountability to the public that bears mission risk and cost. It is also guided by justice across time; shortcuts that create hidden risk today can impose harm on future crews and programs. A responsible threshold therefore combines technical evidence with governance safeguards, including stop/go criteria that cannot be overridden without documented, transparent, and accountable justification.

Appendix:

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Logistic regression model for O-ring damage
logit(P(Damaged=1)) = 5.0686 + (-0.1153) * Temp
Training ROC AUC: 0.6912
Training Log Loss: 0.2188

Requested predicted probabilities:
 Temperature_F  Predicted_Prob_Damage
            50              0.332086
            40              0.611755
            30              0.833156
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